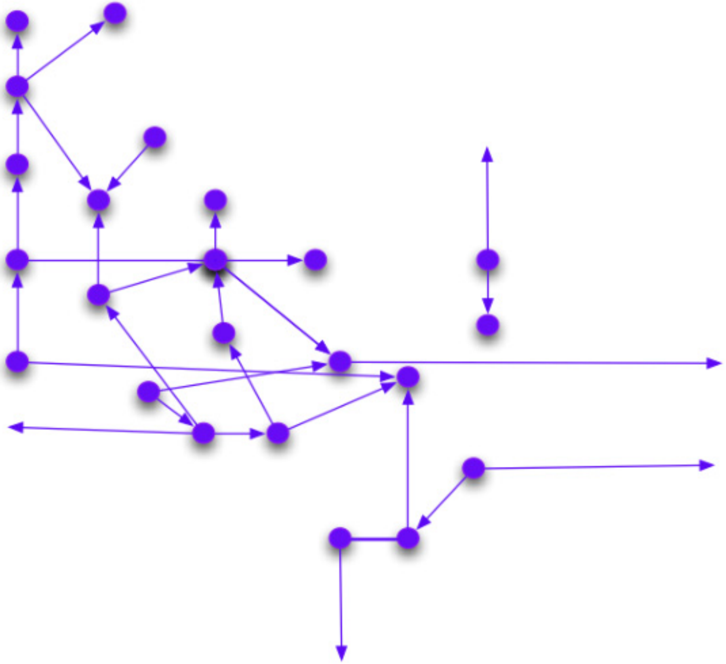
Semantic Web / Web of Data



Why the Semantic Web?…

From `https://www.w3.org/2001/sw/`

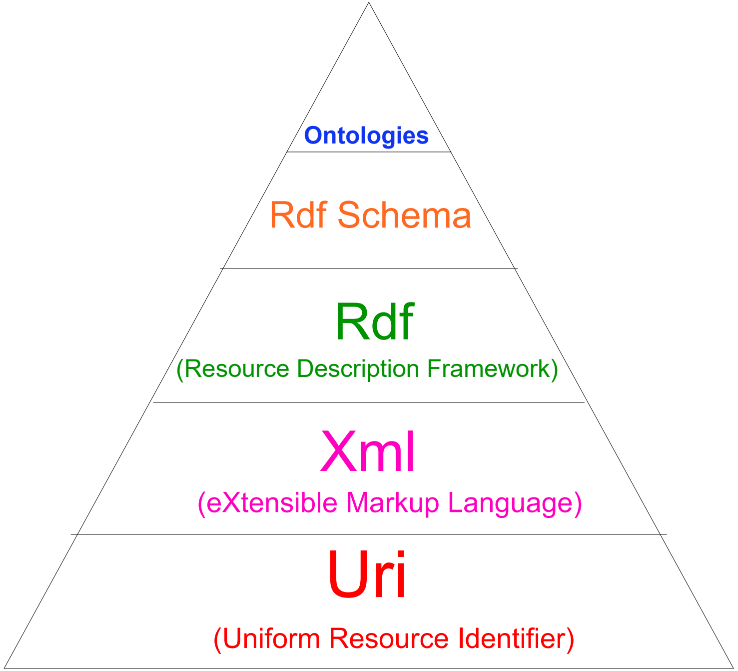
- The Semantic Web is a *Web of data*. There is lots of data we all use every day, and it is not part of the Web. I can see my bank statements on the Web, and my photographs, and I can see my appointments in a calendar. But can I see my photos in a calendar to see what I was doing when I took them? Can I see bank statement lines in a calendar? No!
- Why not? Because we don't have a *Web of data*. Because data is controlled by applications, and each application keeps it to itself

… why the Semantic Web?

- The Semantic Web is about *common formats for integration and combination of data* drawn from diverse sources, where on the original Web mainly concentrated on the interchange of documents

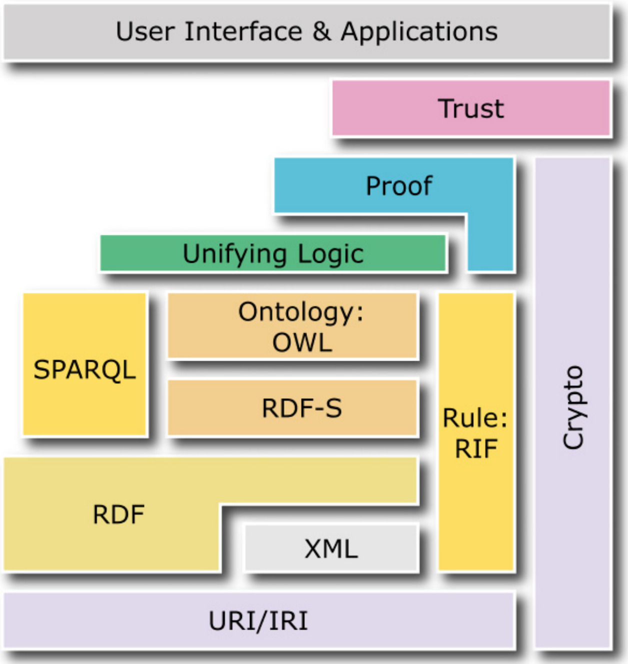
- The Semantic Web is also about languages for recording how the data relates to real-world objects. That allows a person, or a machine, to start off in one database, and then move through an unending set of databases which are connected not by wires but by being about the same thing

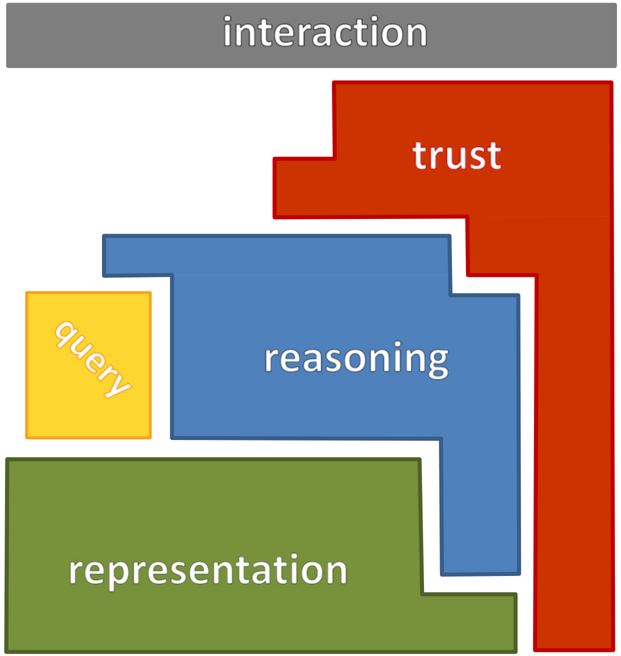
Architecture of the Semantic Web



Semantic Web standard stack

From *a W3C tutorial by Fabien Gandon (INRIA – W3C)*





RDF…

RDF stands for:

- *Resource*: e.g., pages, dogs, ideas... everything that can have a URI
- *Description*: attributes, features, and relations of the resources
- *Framework*: model, languages, and syntaxes for these descriptions

RDF is a “triple model”, i.e., every piece of knowledge is broken down into:

(subject, predicate, object)

… RDF…

Example:

doc.html has for author *Marco Porta* and has for theme *JavaScript*

doc.html has for author Marco Porta
doc.html has for theme JavaScript

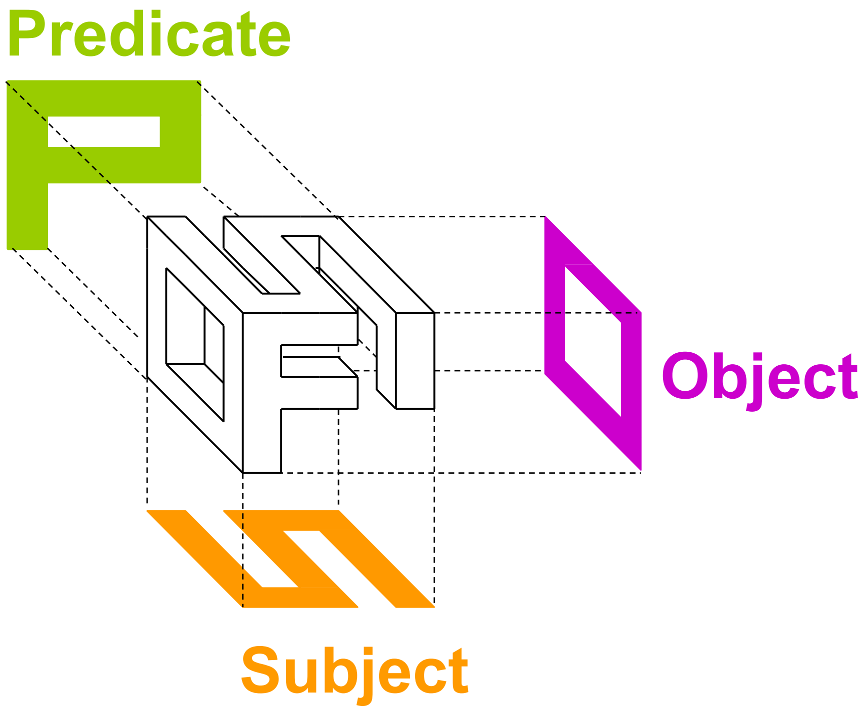
(**subject**, **predicate**, **object**)

(doc.html, author, Marco Porta)

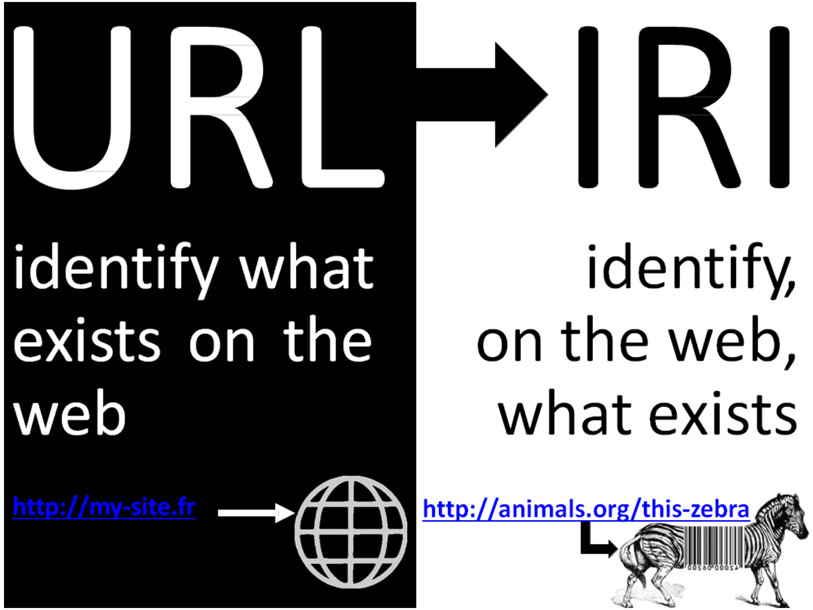
(doc.html, theme, JavaScript)

… RDF

The RDF "atom"



URI/IRI…



URI:

<u>U</u>niform <u>R</u>esource <u>I</u>dentifier

IRI:

<u>I</u>nternational <u>R</u>esource <u>I</u>dentifier: generalization of URI - while URIs are limited to a subset of the ASCII character set, IRIs may contain characters from the Universal Character Set (Unicode/ISO 10646)

… URI/IRI…

A URI can be a URL (Uniform Resource Locator) or a URN (Uniform Resource Name)

URL

- Unique address of a file that is accessible on the Internet

URN

- An Internet resource with a name that, unlike a URL, has persistent significance (i.e., the owner of the URN can expect that someone else, or a program, will always be able to find the resource)

… URI/IRI

Example

Hypothetical URN: `urn:def://blue_laser`, where `def://` might indicate an agency or an accessible directory of all dictionaries, glossaries, and encyclopedias on the Internet, and "blue laser" is the name of a term

A comparable URL would need to specify one specific location for a definition, such as: `https://www.whatis.com/bluelaser.html`

RDF syntax

RDF has an XML syntax

Example:

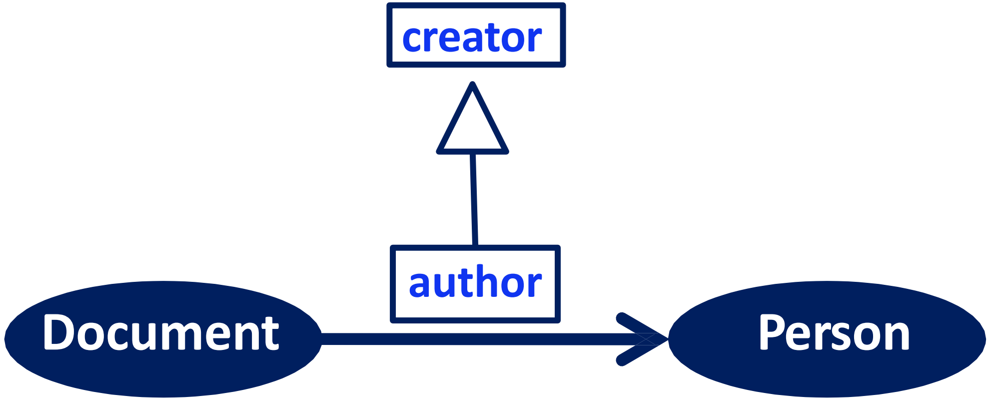
```
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
         xmlns:inria="http://inria.fr/schema#”>
  <rdf:Description rdf:about="http://inria.fr/rr/doc.html">
      <inria:author rdf:resource="http://ns.inria.fr/fabien.gandon#me"/>
      <inria:theme>Music</inria:theme>
  </rdf:Description>
</rdf:RDF>
```

SPARQL is a query language and protocol for RDF

RDFS (RDF Schema)…

Allows to define classes of resources, define relations between resources, and organize their hierarchy

Example:



… RDFS

Example:

```
<rdf:RDF  xml:base ="http://inria.fr/2005/humans.rdfs"
 xmlns:rdf ="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
 xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#"
 xmlns    ="http://www.w3.org/2000/01/rdf-schema#>
 <Class rdf:ID="Man">
   <subClassOf rdf:resource="#Person"/>
   <subClassOf rdf:resource="#Male"/>
   <label xml:lang="en">man</label>
   <comment xml:lang="en">an adult male person</comment>
 </Class>
```

```
<Man> a Class ; subClassOf <Person>, <Male>.
```

OWL (Ontology Web Language)

Knowledge representation language for authoring *ontologies*. Ontologies allow to formally describe *taxonomies,* defining the structure of knowledge in different domains

Example:

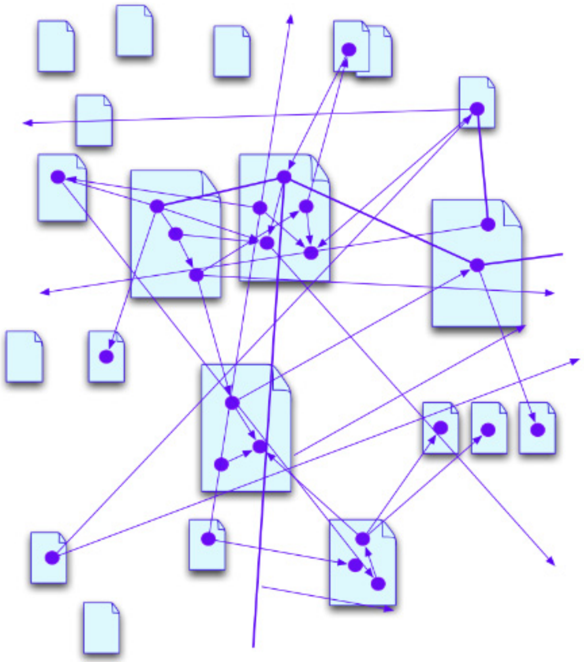
```
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
         xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#"
         xmlns:owl="http://www.w3.org/2002/07/owl#"
         xmlns:dc="http://purl.org/dc/elements/1.1/">

    <!-- OWL Header Example -->
    <owl:Ontology rdf:about="http://www.linkeddatatools.com/plants">
        <dc:title>An Example Ontology</dc:title>
        <dc:description>An example ontology to show its basic syntax</dc:description>
    </owl:Ontology>

    <!-- OWL Class Definition Example -->
    <owl:Class rdf:about="http://www.linkeddatatools.com/plants#planttype">
        <rdfs:label>The plant type</rdfs:label>
        <rdfs:comment>The class of plant types.</rdfs:comment>
    </owl:Class>
</rdf:RDF>
```

Possible applications of the Semantic Web

- Web of trust
- Service discovery
- Home automation
- IoT
- …



Web of Data…

- Focus on practical interoperability

  The Web of Data prioritizes seamless data integration over formal reasoning for real-world applications

- Linked data principles

  URIs and HTTP link resources across datasets and domains

- Lightweight and industry-driven

  Technologies like JSON-LD and RDF facilitate easy adoption with less complexity compared to the Semantic Web

- Broad adoption and applications

  Supports knowledge graphs, IoT ecosystems, and data exchange powering modern web services and AI

`https://www.w3.org/2013/data/`

… Web of Data

Vision vs. Practice: key differences

"Old" Semantic Web: the *Vision*	"New" Web of Data: the *Practice*
A Web for machines to **understand**	A Web for machines to **connect**
• **Focus**: Formal Semantics & Reasoning • **Core Idea**: Artificial Intelligence • **Key Tech**: Ontologies (OWL), Logic • **Analogy**: A global "intelligent" database	• **Focus**: Data Publishing and Connecting • **Core Idea**: Linked Data Principles • **Key Tech**: URIs, HTTP, RDF • **Analogy**: A global "knowledge graph"